11/11/2026

Last time:

$$\frac{1}{4}\cdot\frac{N}{\ln N} < \pi(N) < \frac{13}{4}\cdot\frac{N}{\ln N}, \quad N \geq 2.$$

Arithmetic functions

$$f : \mathbb{N} \longrightarrow \mathbb{R} \text{ or } \mathbb{C}$$

$$\mathbb{N} = \{1, 2, 3, \ldots\}$$

In particular, $\{f(n)\}_{n=1}^{\infty}$ is a sequence.

Eg. 1. $f(n) = 1$, $n \in \mathbb{N}$.

2. $f(n)$ = number of positive divisors of n.

$f(6) = 4$, divisors of 6 are: 1, 2, 3, 6.

3. $f(n)$ = sum of positive divisors of n.

$f(6) = 1 + 2 + 3 + 6 = 12 = 2 \cdot 6$.

4. $\pi(n)$ = number of primes $\leq n$.

5. $f(n) = \begin{cases} 1, & \text{if } n \text{ is prime.} \\ 0, & \text{otherwise.} \end{cases}$

So $\pi(N) = \sum_{n=1}^{N} f(n)$, $f(n)$ as in 5.

6. $f(n) = i^n$.

Viewing f as a sequence, we are interested in

1. $\lim_{n \to \infty} f(n)$

2. asymptotic behavior: how fast does

$f(n)$ converge to its limit,

or fast does it grow or oscillate?

Can also look at its average value

$$\frac{1}{N} \sum_{n=1}^{N} f(n).$$

As an arithmetic function, we look at

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$$

and the relations between

$f(n)$ and $f(p_1^{e_1}), f(p_2^{e_2}), \cdots, f(p_k^{a_k})$.

Two important classes

1. Additive functions.

$$f(mn) = f(m) + f(n), \quad \gcd(m,n) = 1.$$

If

$$f(mn) = f(m) + f(n), \quad \forall\, m, n \geq 1,$$

then f is called completely additive.

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} \xrightarrow{f} f(n) = f(p_1^{e_1}) + f(p_2^{e_2}) + \cdots + f(p_k^{e_k})$$

1. $f(n) = 0$
2. $f(n) = \log_a n$
3. $f(n) = e_p(n)$

} completely additive

4. $f(n) = \omega(n) =$ number of prime factors of n.

$$= \sum_{p|n} 1.$$

e.g. $\omega(54) = 2$, $\omega(2) + \omega(3^3) = 2$.

$54 = 2 \cdot 3^3$, $\omega(2) = 1$, $\omega(3^3) = 1$

but $\omega(4) \neq \omega(2) + \omega(2)$

$4 = 2 \cdot 2$, $\omega(4) = 1$, $\omega(2) = 1$, $\omega(2) = 1$

If $(m, n) = 1$, then any prime p cannot divide both m and n. So

$$\omega(mn) = \sum_{p|mn} 1 = \sum_{p|m} 1 + \sum_{p|n} 1 = \omega(m) + \omega(n).$$

5. Given additive f, g.

$$f(mn) = f(m) + f(n)$$
$$+ \quad + \quad +$$
$$g(mn) = g(m) + g(n)$$

$$\Rightarrow \underbrace{f(mn) + g(mn)}_{(f+g)(mn)} = \underbrace{f(m) + g(m)}_{(f+g)(m)} + \underbrace{f(n) + g(n)}_{(f+g)(n)}$$

Also,

$$\underbrace{cf(mn)}_{(cf)(mn)} = \underbrace{cf(m)}_{(cf)(m)} + \underbrace{cf(n)}_{(cf)(n)}$$

So a linear combination of additive functions is additive.

E.g. $f(n) = \Omega(n) = \sum_{p} e_p(n)$ is completely additive.

$n = p_1^{a_1} \cdots p_k^{a_k}$! $\Omega(n) = a_1 + a_2 + \cdots + a_k$

2. Multiplicative functions

$$f(mn) = f(m)\,f(n) \quad , \quad \gcd(m,n) = 1 .$$

If

$$f(mn) = f(m)\,f(n) \quad , \quad \forall\, m, n \geq 1 .$$

then f is called completely multiplicative.

So

$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$$

$$\downarrow f$$

$$f(n) = f(p_1^{e_1})\, f(p_2^{e_2}) \cdots f(p_k^{e_k}) .$$

Eg. 1. $f(n) = 0$

2. $f(n) = 1$

3. $f(n) = n^c$, $c \in \mathbb{C}$.

4. $f(n) = \tau(n)$ or $d(n)$

$= $ number of positive divisors of n.

Say $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$.

$d \mid n$

$d = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, $0 \le a_i \le e_i$ $(1 \le i \le k)$.

So $\tau(n) = \prod_{i=1}^{k} (1 + e_i)$.

If $(m,n)=1$. $p \mid mn \Rightarrow p \mid m$ or $p \mid n$

$\Downarrow$ but not both

$$\tau(mn) = \prod_{p \mid mn} (1 + e_p(mn))$$

$$= \prod_{p \mid m} (1 + e_p(m)) \prod_{p \mid n} (1 + e_p(n))$$

$$= \tau(m)\,\tau(n).$$

But $\tau(4) \neq \tau(2)\,\tau(2)$

$\tau(4) = 3$, $\tau(2) = 2$, $\tau(2) = 2$